

PADHAISHURU

GATE Preparation Platform

GATE CE

Civil Engineering

TREND-BASED MOCK PAPER

— 03 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	9 subjects

Difficulty Distribution

Easy: 17 questions (26%)

Moderate: 38 questions (58%)

Difficult: 10 questions (15%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE CE — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [-0.33]

[General Aptitude]

Volume of sphere r:

- (A) $\frac{4}{3} \pi r^3$
- (B) $\frac{4}{3} \pi r^2$
- (C) $\frac{4}{3} \pi r$
- (D) $2\pi r^3$

Q2. 1M [no -ve]

[General Aptitude]

Choose similar to 'diligent':

Enter your answer here

Q3. 1M [no -ve]

[Engineering Mathematics]

Order of $\frac{dy}{dx}=y$:

Enter your answer here

Q4. 1M [-0.33]

[Structural Engineering]

Method of joints gives:

- (A) Forces only
- (B) Moments
- (C) Both
- (D) None

Q5. 1M [-0.33]

[Structural Engineering]

Bending moment in UDL simply supported:

- (A) $\frac{wL^2}{8}$
- (B) $\frac{wL^2}{12}$
- (C) $\frac{wL}{2}$
- (D) wL^2

Q6. 1M [-0.33]

[Structural Engineering]

Moment-area method gives:

- (A) Slope
- (B) Deflection
- (C) Both
- (D) None

Q7. 1M [no -ve]

[Geotechnical Engineering]

Bearing capacity of footing:

Enter your answer here

Q8.

1M [-0.33]

[Geotechnical Engineering]

Atterberg limits:

- (A) Liquid limit
- (B) Plastic limit
- (C) Shrinkage
- (D) All

Q9.

1M [-0.33]

[Water Resources Engineering]

Gumbel distribution used for:

- (A) Rainfall
- (B) Flood frequency
- (C) Runoff
- (D) None

Q10.

1M [-0.33]

[Water Resources Engineering]

Manning's equation for:

- (A) Pipe
- (B) Open channel
- (C) Laminar
- (D) Turbulent

Q11.

1M [-0.33]

[Water Resources Engineering]

Darcy-Weisbach for:

- (A) Minor losses
- (B) Major losses
- (C) Both
- (D) None

Q12.

1M [-0.33]

[Water Resources Engineering]

Rainfall measured by:

- (A) Hygrometer
- (B) Rain gauge
- (C) Anemometer
- (D) Barometer

Q13.

1M [-0.33]

[Environmental Engineering]

BOD stands for:

- (A) Biological Oxygen Demand
- (B) Basic Oxygen Demand
- (C) Bacterial Oxygen Demand
- (D) None

Q14.

1M [-0.33]

[Environmental Engineering]

Sedimentation removes:

- (A) Dissolved
- (B) Suspended
- (C) Colloids
- (D) Gases

Q15.

1M [-0.33]

[Transportation Engineering]

PCU stands for:

- (A) A
- (B) B
- (C) C
- (D) D

Q16.

1M [-0.33]

[Transportation Engineering]

Camber provided for:

- (A) Strength
- (B) Drainage
- (C) Smoothness
- (D) Aesthetics

Q17.

1M [-0.33]

[Surveying & Geomatics]

Theodolite measures:

- (A) Distance
- (B) Angle
- (C) Level
- (D) All

Q18.

1M [-0.33]

[Surveying & Geomatics]

Dumpy level used for:

- (A) Angle
- (B) Distance
- (C) Levelling
- (D) All

Q19.

1M [-0.33]

[Construction Materials]

Initial setting time OPC:

- (A) 30 min
- (B) 45 min
- (C) 60 min
- (D) 24 hr

Q20.

1M [no -ve]

[Construction Materials]

Water-cement ratio affects:

*Enter your answer here***Q21.**

1M [-0.33]

[General Aptitude]

Volume of sphere r:

- (A) $\frac{4}{3} \pi r^3$
- (B) $\frac{4}{3} \pi r^2$
- (C) $\frac{4}{3} \pi r$
- (D) $2\pi r$

Q22.

1M [-0.33]

[Environmental Engineering]

BOD stands for:

- (A) Biological Oxygen Demand
- (B) Basic Oxygen Demand
- (C) Bacterial Oxygen Demand
- (D) None

Q23.

1M [no -ve]

[Geotechnical Engineering]

Bearing capacity of footing:

*Enter your answer here***Q24.**

1M [-0.33]

[Surveying & Geomatics]

Dumpy level used for:

- (A) Angle
- (B) Distance
- (C) Levelling
- (D) All

Q25.

1M [-0.33]

[Geotechnical Engineering]

Bearing capacity of footing:

- (A) Depends on soil
- (B) Depends on load
- (C) Both
- (D) None

Q26.

1M [no -ve]

[Surveying & Geomatics]

Theodolite measures:

Enter your answer here

Q27.

1M [-0.33]

[Surveying & Geomatics]

Theodolite measures:

- (A) Distance
- (B) Angle
- (C) Level
- (D) All

Q28.

1M [no -ve]

[Water Resources Engineering]

Manning's equation for:

*Enter your answer here***Q29.**

1M [-0.33]

[Water Resources Engineering]

Rainfall measured by:

- (A) Hygrometer
- (B) Rain gauge
- (C) Anemometer
- (D) Barometer

Q30.

1M [-0.33]

[Transportation Engineering]

Camber provided for:

- (A) Strength
- (B) Drainage
- (C) Smoothness
- (D) Aesthetics

Q31.

2M [-0.66]

[Construction Materials]

Initial setting time OPC:

- (A) 30 min
- (B) 45 min
- (C) 60 min
- (D) 24 hr

Q32.

2M [no -ve]

[Geotechnical Engineering]

Atterberg limits:

*Enter your answer here***Q33.**

2M [-0.66]

[Surveying & Geomatics]

Dumpy level used for:

- (A) Angle
- (B) Distance
- (C) Levelling
- (D) All

Q34.

2M [no -ve]

[Geotechnical Engineering]

Atterberg limits:

Enter your answer here

Q35.

2M [-0.66]

[Structural Engineering]

Bending moment in UDL simply supported:

- (A) $wL^2/8$
- (B) $wL^2/12$
- (C) $wL/2$
- (D) wL^2

Q36.

2M [no -ve]

[Water Resources Engineering]

Rainfall measured by:

Enter your answer here

Q37.

2M [no -ve]

[Water Resources Engineering]

Darcy-Weisbach for:

Enter your answer here

Q38.

2M [no -ve]

[Structural Engineering]

Method of joints gives:

Enter your answer here

Q39.

2M [-0.66]

[Water Resources Engineering]

Darcy-Weisbach for:

- (A) Minor losses
- (B) Major losses
- (C) Both
- (D) None

Q40.

2M [no -ve]

[Geotechnical Engineering]

Atterberg limits:

Enter your answer here

Q41.

2M [-0.66]

[Water Resources Engineering]

Darcy-Weisbach for:

- (A) Minor losses
- (B) Major losses
- (C) Both
- (D) None

Q42.

2M [-0.66]

[Environmental Engineering]

Sedimentation removes:

- (A) Dissolved
- (B) Suspended
- (C) Colloids
- (D) Gases

Q43.

2M [no -ve]

[Construction Materials]

Water-cement ratio affects:

*Enter your answer here***Q44.**

2M [no -ve]

[Structural Engineering]

Moment-area method gives:

- (A) Slope
- (B) Deflection
- (C) Both
- (D) None

Q45.

2M [-0.66]

[Construction Materials]

Water-cement ratio affects:

- (A) Strength only
- (B) Workability only
- (C) Both
- (D) None

Q46.

2M [-0.66]

[Structural Engineering]

Moment-area method gives:

- (A) Slope
- (B) Deflection
- (C) Both
- (D) None

Q47.

2M [no -ve]

[Geotechnical Engineering]

Atterberg limits:

*Enter your answer here***Q48.**

2M [-0.66]

[Structural Engineering]

Moment-area method gives:

- (A) Slope
- (B) Deflection
- (C) Both
- (D) None

Q49.

2M [no -ve]

[Surveying & Geomatics]

Theodolite measures:

*Enter your answer here***Q50.**

2M [-0.66]

[General Aptitude]

Volume of sphere r:

- (A) $\frac{4}{3} \pi r^3$
- (B) $\frac{4}{3} \pi r^2$
- (C) $\frac{4}{3} \pi r$
- (D) $2\pi r^2$

Q51.

2M [-0.66]

[Environmental Engineering]

Sedimentation removes:

- (A) Dissolved
- (B) Suspended
- (C) Colloids
- (D) Gases

Q52.

2M [-0.66]

[General Aptitude]

Choose similar to 'diligent':

- (A) Lazy
- (B) Industrious
- (C) Careless
- (D) Ignorant

Q53.

2M [no -ve]

*[Engineering Mathematics]*Order of $dy/dx=y$:

*Enter your answer here***Q54.**

2M [-0.66]

[Construction Materials]

Water-cement ratio affects:

- (A) Strength only
- (B) Workability only
- (C) Both
- (D) None

Q55.

2M [no -ve]

[Surveying & Geomatics]

Dumpy level used for:

- (A) Angle
- (B) Distance
- (C) Levelling
- (D) All

Q56.

2M [no -ve]

[Geotechnical Engineering]

Bearing capacity of footing:

*Enter your answer here***Q57.**

2M [-0.66]

[Water Resources Engineering]

Rainfall measured by:

- (A) Hygrometer
- (B) Rain gauge
- (C) Anemometer
- (D) Barometer

Q58.

2M [-0.66]

[Construction Materials]

Water-cement ratio affects:

- (A) Strength only
- (B) Workability only
- (C) Both
- (D) None

Q59.

2M [-0.66]

[Surveying & Geomatics]

Dumpy level used for:

- (A) Angle
- (B) Distance
- (C) Levelling
- (D) All

Q60.

2M [-0.66]

[Structural Engineering]

Moment-area method gives:

- (A) Slope
- (B) Deflection
- (C) Both
- (D) None

Q61.

2M [-0.66]

[Construction Materials]

Water-cement ratio affects:

- (A) Strength only
- (B) Workability only
- (C) Both
- (D) None

Q62.

2M [no -ve]

[General Aptitude]

Volume of sphere r:

- (A) $\frac{4}{3} \pi r^3$
- (B) $\frac{4}{3} \pi r^2$
- (C) $\frac{4}{3} \pi r$
- (D) $2\pi r$

Q63.

2M [-0.66]

[Surveying & Geomatics]

Theodolite measures:

- (A) Distance
- (B) Angle
- (C) Level
- (D) All

Q64.

2M [-0.66]

*[Engineering Mathematics]*Order of $dy/dx=y$:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q65.

2M [no -ve]

[Transportation Engineering]

PCU stands for:

Enter your answer here